

NBA opening-spread walk-forward — model and strategy review

Code, data pipeline and the full research register: github.com/sean-qin-usa/nba-prediction-model

Strategy

Directional value-taking against the opening point spread in NBA regular-season games. A market-blind margin model prices every listed game; its disagreement with the opening spread is then spent, at about a third of face value, as a correction to the opening spread. The architecture and its transformations are fixed ex ante. Within each walk-forward fold the team ratings, availability probabilities, schedule effects and offset coefficients are estimated exclusively from prior seasons:

$$m_{\text{offset}} = m_{\text{open}} + f(m_{\text{blind}} - m_{\text{open}}, \text{rest differential}, |m_{\text{open}}|)$$

where f is a ridge regression shrunk hard toward zero and refitted in each fold. Its coefficient on the fundamental disagreement $m_{\text{blind}} - m_{\text{open}}$ lies in $[0.33, 0.37]$ in every fold, so the model's stated edge is spent at roughly a third of face value. A single side is taken when the corrected margin still disagrees with the line by more than a walk-forward selected threshold, at the best number available across books. Every position is one side of one game at -110 , entered at the open and held to settlement, with no hedge, offsetting leg, in-game management or exit. Each bet is a flat 1 unit, so returns do not compound, and no calendar filter is applied.

Selection is the strategy. The book below is 460 bets out of roughly 3,690 regular-season games in the window, about 12% of the slate. A side is taken only where the corrected margin disagrees with the opening spread by more than the walk-forward threshold, so the cover rate on the selected book is not the cover rate on the slate.

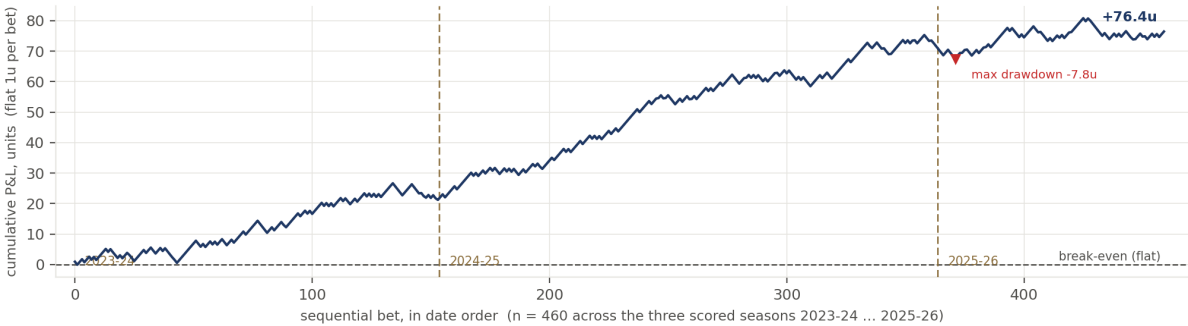
Simulation: 460 bets, 3 seasons, 2023-24 → 2025-26. The configuration is selected on seasons 1..k from a pre-declared space, frozen, scored on season k+1, and rolled forward; nothing is refitted on the season being scored. The selection history reaches back to 2012-13, so the three seasons here are scored against a decade of prior data rather than against each other. Results are reported at a modelled best-of-nine execution tier. That tier is observed directly in 2023-24, where the panel averages 7.74 books per game, and inferred from the measured shopping relationship in 2024-25 and 2025-26.

Headline results

season	books/game at open	bets	P&L (u)	ROI	Sharpe (ann.)
2023-24	7.74	154	+21.20u	+13.77%	1.90
2024-25	1.00	210	+50.40u	+24.00%	3.81
2025-26	1.03	96	+4.85u	+5.05%	0.54
2023-26 pooled — recent execution headline	—	460	+76.44u	+16.62%	2.25
2019-26 — primary full-frame reference	—	888	+80.93u	+9.11%	1.10

ROI is P&L per unit staked; Sharpe is annualized on realised sessions. The multi-book panel is measured only in 2023-24, so the pooled figures are modelled for 2024-25 and 2025-26. The final row reports the full seven-season frame and serves as the primary reference; the three-season headline is the recent portion of that same sample.

Best-of-nine is the most consequential execution assumption in the headline. At a single observed book the same bets return +10.63%, so approximately two-thirds of the reported ROI survives with no line shopping at all; the full ladder appears under Execution below.



All 460 bets in date order at the opening spread and -110 ; flat 1u, so the path is a running sum and nothing is re-selected within it. Max drawdown is $-7.75u$ at bet 372. Over the full seven-season frame the path is $+80.93u$ on 888 bets with a $-23.1u$ drawdown, the first three of those seasons losing money and the last four making $+83.5u$.

Pooled ROI is $+16.62\%$ over 460 bets, with a season-clustered 95% interval of $[-9.29\%, +37.83\%]$; on the full seven-season frame it is $+9.11\%$ with an interval of $[-2.75\%, +16.08\%]$. Three seasons is a small sample and both intervals contain zero. 2024-25 supplies 66% of the three-season P&L.

The model

The model outputs a margin, not a probability. A spread is itself a margin forecast, so the sides test involves no devig or probability conversion.

```
margin = 0.5·four_factors + 0.5·availability_composition + schedule_layer + tank_term
P(home win) = sigmoid(margin / 7.2)
```

Two independent estimates of team strength, averaged, plus additive context. The market-blind model never sees market odds, enforced structurally rather than by convention. Only the offset layer above it sees the price.

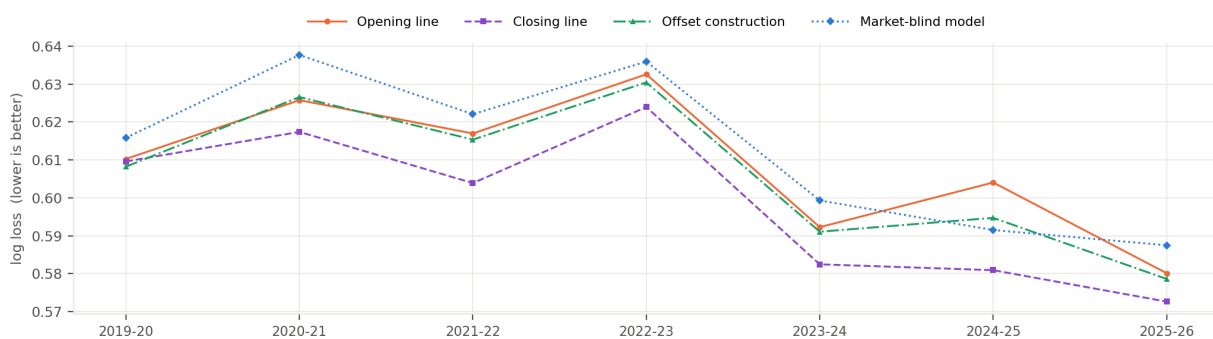
component	what it is	how it is fit
Four factors	opponent-adjusted ratings on shooting, turnovers, rebounding and free-throw rate; a decomposition of points per possession rather than a feature shortlist	one L2 ridge solve per factor (ridge=25), then the four adjusted values mapped to points by a fitted linear map. An 8-factor extension tied, so 4 is kept for parsimony.
Availability composition	Σ over available players of DARKO talent x trailing minutes / 48; the leg that reacts to injuries	each player weighted by $1 - P(\text{out})$, forecast from information available at the open. Requires the daily injury report; this is why the model frame starts 2019-20
Schedule layer	home edge, back-to-backs, dead-team flags	estimated walk-forward, EB shrinkage $n / (n+600)$ toward a prior, team_home_ridge=200 on per-team home deviations. The only component that has survived strict out-of-sample testing on every split tried.
Tank term	late-season effort	exactly zero outside its window
Offset layer	the correction applied to the opening line	ridge on three features knowable at the open, shrunk hard toward zero. Fitted edge coefficient 0.33-0.37 in every fold

Regularisation, in four forms: L2 ridge on the ratings solve; empirical-Bayes shrinkage toward a prior in the schedule and tank layers; data augmentation via prior-season pseudo-observations; and Bayesian/EB priors in the usage and props fits. The link scale 7.2 is the frozen production constant. It is a plug-in rather than a tuned one — matching a logistic to the margin-residual normal gives 7.53 from training residuals alone, and re-deriving it inside each fold is worth about 0.0002 nats, not significant. Every log-loss comparison in this document recalibrates each source walk-forward on prior seasons, so no forecast is advantaged by another's scale.

Inputs are four, and all public: nba_api box scores, the daily NBA injury report (PDF ingest), DARKO talent ratings, and historical odds, the last used by the offset layer and for scoring but never by the market-blind model. The entire purchasable stack — professional minutes projections, tracking feeds and premium talent ratings — improves log loss by 0.0012 nats combined, an effect whose interval contains zero. Fitting is restricted to the 002 prefix, 35,546 regular-season games.

Player-, lineup- and matchup-level alternatives were built and gated rather than skipped — defensive RAPM, defended-FG% by shot category, Synergy play-types, a player skill posterior and 151,914 lineup stints among them. None improved significantly on the shipped DARKO-based availability composition, which is the one player-level term that survived. The full inventory and each gate result are in the repository.

Accuracy. All four forecasts are scored on identical games over the fully-covered frame (2019-20 → 2025-26, K=7, n=8,286), each converted to a probability with its own walk-forward scale.



Log loss by season, all four forecasts on identical games. Lower is better.

pooled, 2019-26 (n=8,286)	opening line	closing line	offset construction	market-blind model
log loss	0.6084	0.5980	0.6059	0.6122

The offset construction improves on the opening line in six of the seven scored seasons and recovers 24.1% of the information the market adds between the open and the close. The market-blind model does not improve on the opener, which is why the strategy is constructed as a correction to the line rather than as a forecast compared against it. Against the closing line its normalized gap — $(ll_{us} - ll_{mkt}) / (ln2 - ll_{mkt})$, the share of the closing line's skill-above-a-coinflip left uncaptured — is 13.59%.

Closing-line value is positive and significant across 19 seasons, so the disagreement the offset layer acts on does carry information relative to the market's final price.

Considerations

Reporting frames

Two frames, each defined by what data exists. **Model accuracy:** 2019-20 → 2025-26 — the daily injury report the availability leg is built on begins 2018-12-17, so 2019-20 is the first fully covered season. **Betting:** the headline covers 2023-24 through 2025-26, the recent execution-study window. Multi-book prices are observed directly in 2023-24 and inferred in the two seasons after it, and the complete seven-season result is reported alongside it in the same table.

Execution

Model held fixed and only the transacted price varied, on the 2023-26 book.

execution	2023-26 ROI	status
1 retail book	+10.63%	single-book baseline
2 books	+12.91%	measured in 2023-24 only
5 books	+15.31%	firm baseline
9 books	+16.62%	max books observed, the reported tier
exchange, 2% commission	+14.72%	arithmetic, no exchange data held

The ladder flattens fast: 36% of the time two books post the same number, so extra books duplicate rather than add.

Methodology

- **Walk-forward selection.** The specification applied to season $k+1$ is chosen from a pre-declared space using seasons $1\dots k$ only, and is scored on $k+1$ alone.
- **Settlement-only fills.** Bets are entered at the recorded opening spread, priced at -110 and held to settlement; the simulation models no queue, partial fill or re-pricing.
- **Bet-time information only.** Availability is constructed from the last injury report published *before* game day, excluding both the 5PM report and the pregame inactive list, which post after the opening line.
- **Regular season only.** The sample is restricted to games carrying the 002 prefix. NBA Cup group-stage games are retained because they count in the standings, and a pre-registered difference-in-differences found no detectable effect on them (MDE 2.21 points).
- **Season-clustered inference,** checked across rolling-origin, leave-one-season-out, block-bootstrap and legacy splits, corrected for multiple comparisons across the running family.

Caveats

- All reported results are simulated. The model is not in production and no capital has been committed.
- Best-of-nine pricing is observed in one season and inferred in two. A measured multi-book panel exists for 2023-24 alone, at 7.74 books per game, while 2024-25 and 2025-26 observe 1.00 and 1.03 respectively, so their prices come from a shopping relationship applied to a single recorded book.
- Best-of- N execution assumes every bet is placed at the most favourable quoted line. Those quotes are also the most exposed to limits or withdrawal: 8.1% sit more than 1.5 points off the next-best book. Charging for them costs approximately 2.3 ROI points, and that charge captures none of the staking limits or account restrictions that arrive sooner in practice.
- 2026-27 is the first genuinely out-of-sample season. The betting configuration is re-selected walk-forward, but the model architecture was chosen on a 2021-26 corpus and passed to every step as fixed.
- Exchange fees, state taxes and account limits are not modelled. Full method, gates and the register of rejected work are in the repository.

Next steps

- **Capture at least two books at the open, beginning opening night.** Best-of-two raises CLV by roughly 49%, and taking the worse of the two erases nearly all of that gain. The logger has never run in-season, so if it is not live on opening night the season's open-price CLV record is lost and cannot be reconstructed afterwards.
- **Run both arms prospectively in 2026-27.** The offset construction is the primary and the market-blind model its live control. They share 65% of their bets and take the same side on every overlapping game, which makes the pairing a controlled comparison rather than a diversified portfolio.
- **Use CLV as the primary live diagnostic.** It is positive in 17 of 19 seasons, requires no de-vigging convention, and resolves within weeks, whereas realised ROI at this dispersion requires far more data than a season provides.
- **Evaluate exchange access if it becomes available.** It is the largest cost lever measured in this project and requires no further modelling change.
- **Prioritise the props engine over further sides research.** Both shipped improvements of the last development cycle came from props, and those markets appear less efficient.

Explicitly not next: further feature search on the sides model. The question the project turns on is whether live CLV against opening prices, with two books captured, reproduces the backtested relationship.